

Program 1: Write a program to create a Calculator.

Code:

```
# Program 1: Basic Calculator

def calculator():

    print("----- Basic Calculator -----")

    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")

    choice = int(input("Enter your choice (1-4): "))

    if choice in [1, 2, 3, 4]:

        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))

        if choice == 1:

            print("Result =", num1 + num2)

        elif choice == 2:

            print("Result =", num1 - num2)

        elif choice == 3:

            print("Result =", num1 * num2)

        elif choice == 4:

            if num2 != 0:

                print("Result =", num1 / num2)

            else:

                print("Error: Cannot divide by zero!")

    else:

        print("Invalid choice! Please select 1-4.")

if __name__ == "__main__":

    calculator()
```

Output:

----- Basic Calculator -----

1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter your choice (1-4): 4

Enter first number: 80

Enter second number: 5

Result = 16.0

Program 2: Write a program where the computer **generate a random number**, and the user tries to guess it.

Code:

```
# Program 2: Random Number Guessing Game
```

```
import random
```

```
def guessing_game():
```

```
    print("----- Random Number Guessing Game -----")
```

```
    secret_number = random.randint(1, 100)
```

```
    attempts = 0
```

```
    while True:
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```
        guess = int(input("Guess a number between 1 and 100: "))
```

```
        attempts += 1
```

```
    if guess < secret_number:
        print("Too Low! Try again.")
    elif guess > secret_number:
        print("Too High! Try again.")
    else:
        print("Congratulations! You guessed it correctly.")
        print("Total attempts:", attempts)
        break
if __name__ == "__main__":
    guessing_game()
```

Output:

```
----- Random Number Guessing Game -----
Guess a number between 1 and 100: 80
Too High! Try again.
Guess a number between 1 and 100: 83
Too High! Try again.
Guess a number between 1 and 100: 60
Too High! Try again.
Guess a number between 1 and 100: 50
Too High! Try again.
Guess a number between 1 and 100: 40
Too High! Try again.
Guess a number between 1 and 100: 25
Congratulations! You guessed it correctly.
Total attempts: 6
```

Program3: Write the ciphertext output to another file:

Code:

```
# Program 3: Caesar Cipher – Write Ciphertext to Another File
```

```
def encrypt_caesar(text, shift):
    encrypted = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            encrypted += chr((ord(ch) - base + shift) % 26 + base)
        else:
            encrypted += ch
    return encrypted

# Read from input file
with open("input.txt", "r") as infile:
    plaintext = infile.read()

# Encrypt plaintext
shift = 3 # you can change shift for different encryption
ciphertext = encrypt_caesar(plaintext, shift)

# Write ciphertext to output file
with open("output.txt", "w") as outfile:
    outfile.write(ciphertext)

print("Ciphertext written successfully to output.txt")
```

Output:

Ciphertext written successfully to output.txt

Input Text: Hello World! This is a test message for Caesar cipher encryption.

**Output Text: Khood Zruog! Wklv lv d whvw phvvdjh iru Fdhvdu flskhu
hqfubswlrq.**